

Q1	A	B	C	D	E	F
Q2	A	B	C	D	E	F
Q3	A	B	C	D	E	F
Q4	A	B	C	D	E	F
Q5	A	B	C	D	E	F
Q6	A	B	C	D	E	F
Q7	A	B	C	D	E	F
Q8	A	B	C	D	E	F
Q9	A	B	C	D	E	F
Q10	A	B	C	D	E	F

1. A crowbar used to pull a nail out of wood is an example of which simple machine?

- A. A screw
- B. A lever
- C. A gear
- D. An inclined plane
- E. A pulley
- F. A wheel and axle

2. When building a robotic hand using cardboard, the fold or crease in the cardboard finger acts as which part of a lever?

- A. The pulley
- B. The effort
- C. The string
- D. The wedge
- E. The load
- F. The fulcrum

3. A student pushes a box with a force of 10 Newtons for a distance of 5 meters. How much work is done?

- A. 5 Joules
- B. 50 Joules
- C. 100 Joules
- D. 15 Joules
- E. 5 Joules
- F. 2 Joules

4. Why are sports cars designed with a wedge shape at the front?

- A. To use larger gears.
- B. To look like a doorstop.
- C. To increase the weight of the car.
- D. To lift the car off the ground.
- E. To cut through air particles (aerodynamics).
- F. To carry more passengers.

5. Why do elevators (lifts) often use a counterweight?

- A. To increase the distance the elevator travels.
- B. To make the elevator look modern.
- C. To make the elevator fall faster.
- D. To increase the friction on the cables.

- E. To act as a wedge if the cable snaps.
- F. To help balance the load so the motor requires less force.
6. If you use a pulley system that reduces the force needed by half, what happens to the rope?
- A. You need to pull the rope a shorter distance.
- B. You need to pull the rope a longer distance.
- C. You need a thicker rope.
- D. The rope ties itself in a knot.
- E. You need a stronger rope.
- F. The rope becomes heavier.
7. A drill bit is a type of screw. It changes turning force into which type of motion?
- A. Circular motion only
- B. Backward motion
- C. Random motion
- D. Linear (forward) motion
- E. Sideways motion
- F. No motion
8. What is the standard unit of measurement for "Work"?
- A. Seconds
- B. Joules
- C. Newtons
- D. Watts
- E. Grams
- F. Meters
9. Mountain roads often use switchbacks (zig-zags) to drive up the mountain. What does this achieve?
- A. It makes the mountain shorter.
- B. It creates a wedge shape.
- C. It creates a gentler slope (shallow angle) requiring less force.
- D. It increases the speed of the car.
- E. It allows the car to use a smaller gear.
- F. It acts as a lever for the car.
10. A box is pushed up a 5-meter long inclined plane with a force of 40 Newtons. How much work is done?
- A. 8 Joules
- B. 45 Joules
- C. 400 Joules
- D. 35 Joules
- E. 200 Joules
- F. 2000 Joules